ChatGPT

# Computing the rate $a_2$ and problem setup

The given code defines $a_2$ as a “computable” real: computing $a_2(n)=2f(n)+\tfrac12+0.25^n$ for increasing $n$ yields a sequence converging to the true $a_2$ [1]. In fact one finds $$f(n)=72^{-E}+\sum_{k=3}^{n+2}(2^{-k}-72^{-E k}),\quad E=10^{24},$$ so as $n\to\infty$, $f(\infty)=\sum_{k=3}^\infty2^{-k}=1/4$ (the tiny $72^{-E}$-terms are negligible). Hence $$a_2=2f(\infty)+\tfrac12=2\cdot\tfrac14+\tfrac12=1.$$ More precisely the $72^{-E}$ terms make $a_2=1+O(72^{-E})$, i.e. slightly above 1. Thus effectively $a_2\approx1.00$ to any practical precision.

The dosage constraint is $a_1x_1+a_2x_2=1$ with $a_1=1$. We minimize total time $T=x_1+x_2$ subject to $x_1+a_2x_2=1$, $x_i\ge0$. This is a linear program whose optimum occurs at a corner of the feasible segment [2]. The two endpoints are $(x_1,x_2)=(1,0)$ or $(0,1/a_2)$. At $(1,0)$ the total time is $T=1$. At $(0,1/a_2)$ the total time is $T=1/a_2$. Since $a_2>1$, we have $1/a_2<1$, so the minimum is achieved by using only the second therapy (setting $x_1=0$).

## Optimal solution and numeric values

Hence the optimal solution is $$x_1=0,\qquad x_2=\frac1{a_2}.$$ Numerically $a_2\approx1.000\dots$, so $1/a_2\approx0.999\dots$. Rounding to two-digit accuracy (in the $\ell^\infty$ sense) gives $$x_1\approx0.00,\quad x_2\approx1.00.$$ These values ensure the total dose $1\cdot x_1+a_2x_2\approx1$ and minimize $x_1+x_2$.

**Answer:** $x_1\approx0.00$, $x_2\approx1.00$.

**Explanation:** The code shows $a_2$ is a computable real very close to 1 [1]. For a linear constraint $x_1+a_2x_2=1$ the minimum of $x_1+x_2$ occurs at an endpoint of the feasible segment [2]. Checking $(1,0)$ vs. $(0,1/a_2)$ shows the latter has smaller total time because $a_2>1$. Thus $x_1=0$, $x_2=1/a_2\approx1$, to two-decimal accuracy $x_1=0.00,\;x_2=1.00$.

**Sources:** Code analysis and linear-programming principle [1] [2].

---


[1] Computable number - Wikipedia
https://en.wikipedia.org/wiki/Computable_number

[2] Problem 34 Consider the objective function ... [FREE SOLUTION] | Vaia
https://www.vaia.com/en-us/textbooks/math/precalculus-6-edition/chapter-7/problem-34-consider-the-objective-function-za-xb-ya0-and-b0-/